Ch 12.4: Single Linkage Clustering

Lecture 28 - CMSE 381

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Wed, Apr 27, 2022

Announcements

Last time:

- Unsupervised learning
- Clustering

This lecture:

- Single Linkage Clustering
- Dendrograms

Announcements:

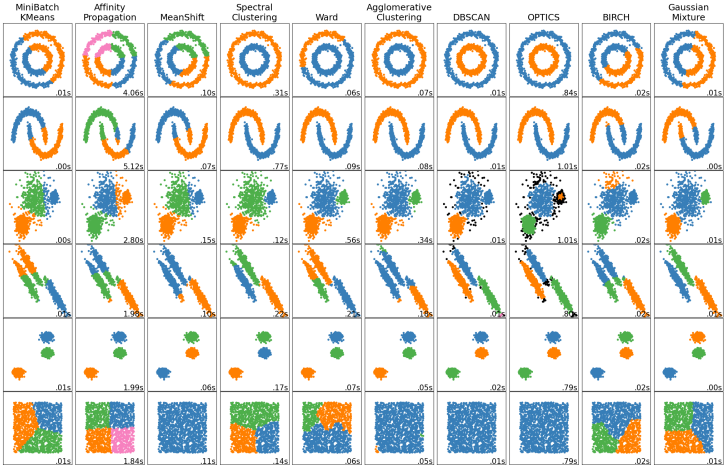
- ~~Mon 4/25: 12.1, 12.4: Clustering~~
- Weds 4/27: Finish 12.4 & Mini lab
  - HW #10 Due Weds
- Fri 4/29: Review - Bring questions!
- Final: May 4, 5:45 - 7:45 PM, Erickson 226
  - Content since midterm (Ch 6 and on)
  - One page (8.5x11) handwritten cheat sheet

Section 1

Last Time: K-Means Clustering

Big idea

Find subgroups/clusters in a data set



Section 2

K-Means Clustering

K-Means Clustering

Pick K in advance

Write sets as $C_1, \cdots, C_K$

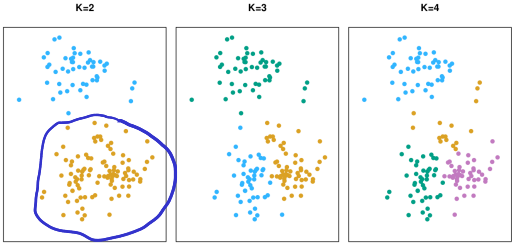
- $C_1 \cup C_2 \cup \cdots C_K = \{1, \cdots, n\}$
- $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$

Goal:

- $\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} W(C_k) \right\}$
- Squared euclidean distance

internal variance

$$W(C_k) = \underbrace{\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \underbrace{(x_{ij} - x_{i'j})^2}}_{\text{variance}}$$

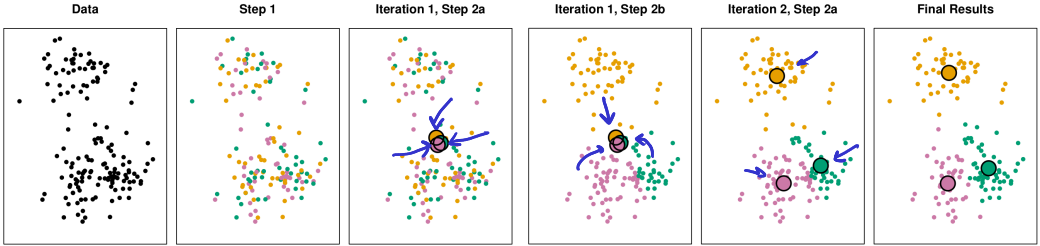


The algorithm

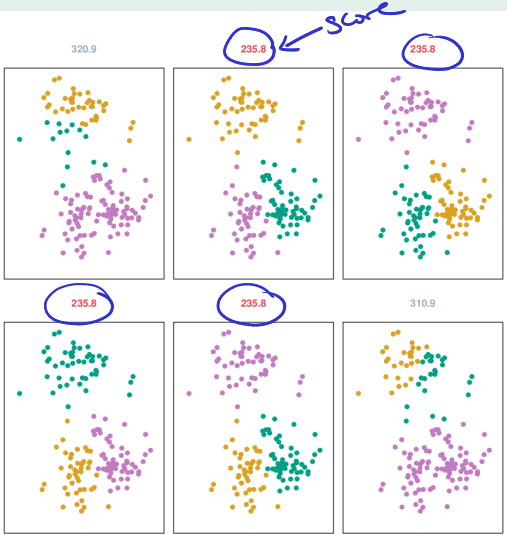
$$\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}$$

Algorithm 12.2 *K-Means Clustering*

1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
   (a) For each of the K clusters, compute the cluster *centroid*. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
   (b) Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).



Dependence on initial conditions

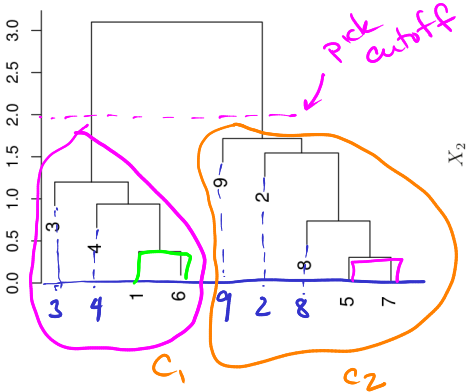


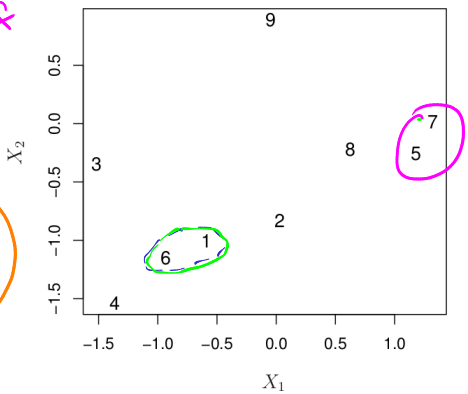
Section 3

Hierarchical Clustering

- No fixed choice of K to start
- No randomness

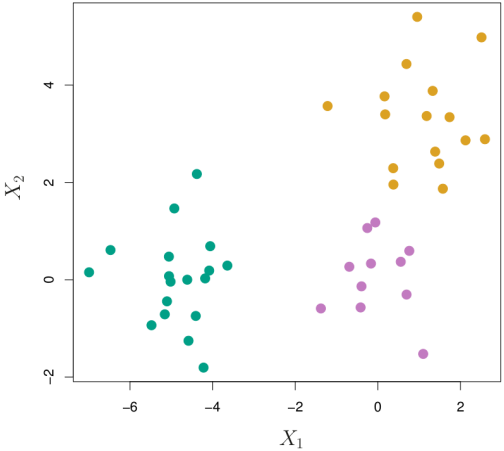
Dendrogram

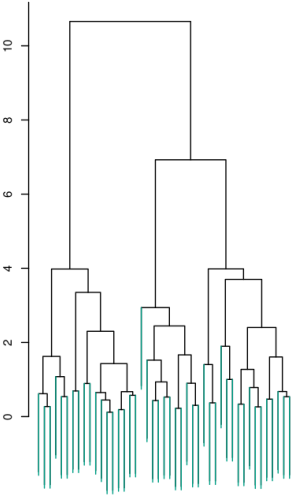




Important info: vertical

A bigger example





Single linkage

2 way to determine similarity

Distance between cluster A and cluster B:
Smallest distance between the points

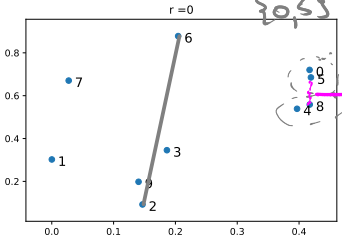
$$L(A, B) = \min_{a \in A, b \in B} \|a - b\|$$

Euclidean distance

$\|a-b\| = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2}$

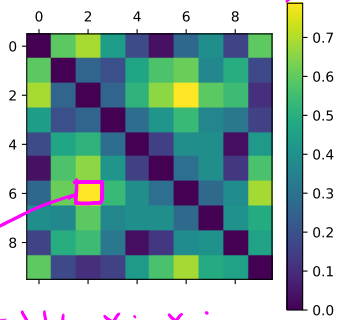
Cluster 1 $\{0,5\}$

Cluster 2 $\{4,8\}$



$L(A,B)$ length

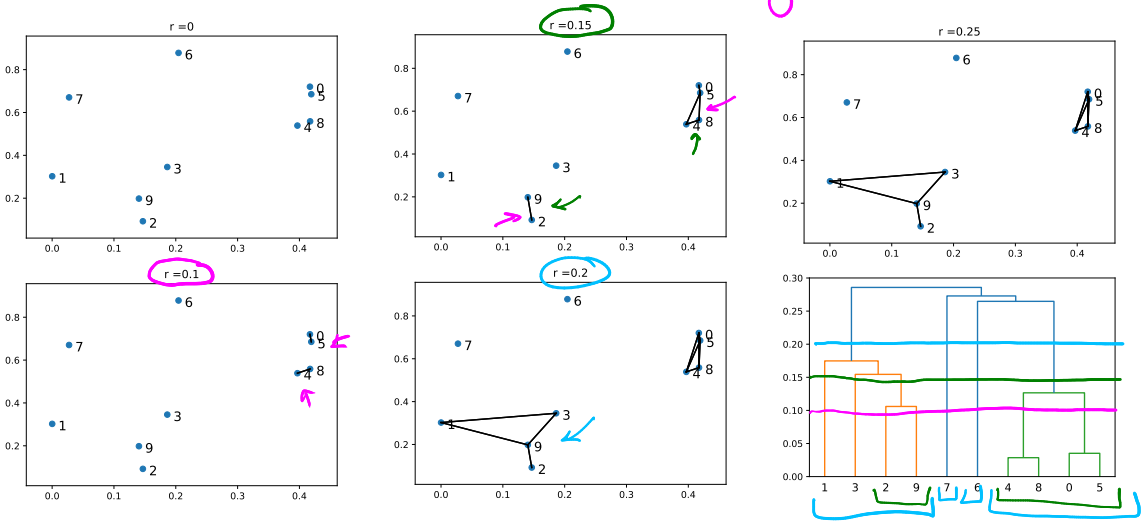
Distance matrix



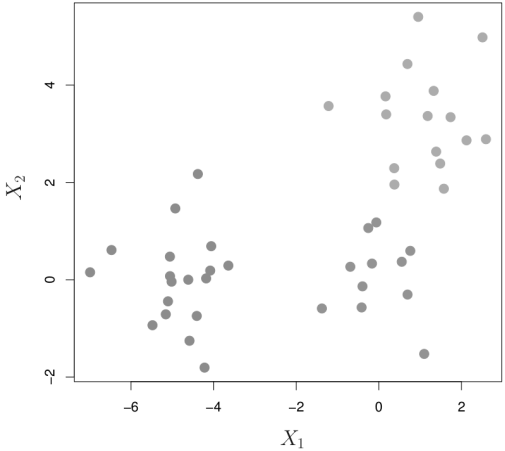
$D[i,j]$ = Dist b/t x_i, x_j

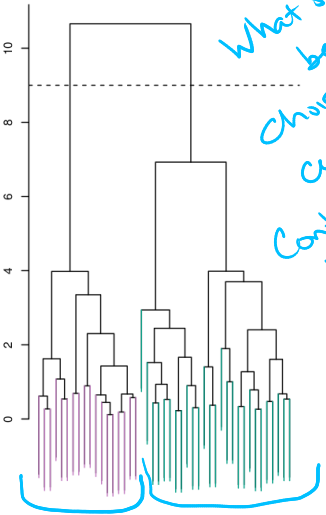
Building the dendrogram

Single Linkage



How to get clusters

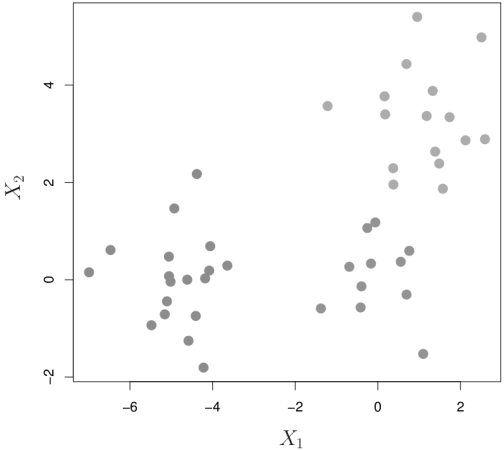


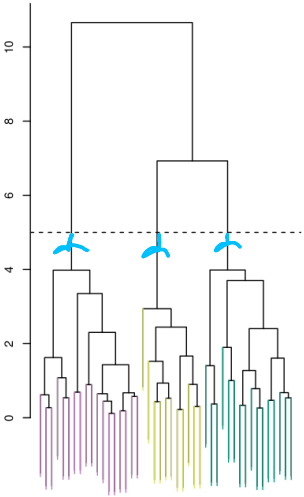


What's the best choice of cutoff?

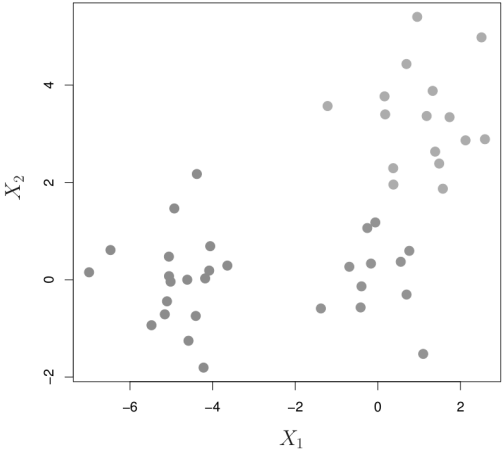
Context dependent

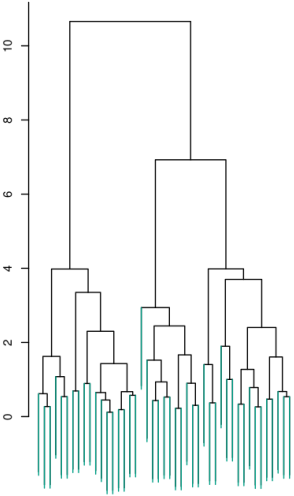
How to get different clusters





Can get any number of clusters



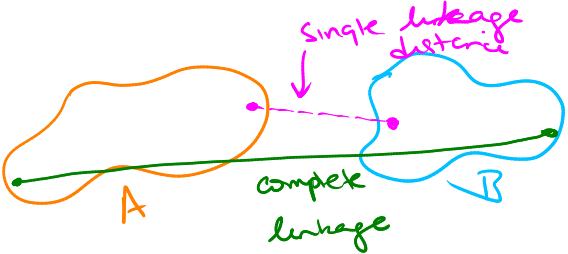


Linkage

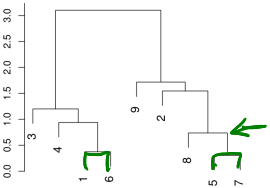
Linkage	*Description*
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *largest* of these dissimilarities.
Single	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *smallest* of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time.
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *average* of these dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable *inversions*.

$$L(A,B) = \max_{\substack{a \in A \\ b \in B}} \|a-b\|$$

$$L(A,B) = \min_{\substack{a \in A \\ b \in B}} \|a-b\|$$



Example with complete linkage

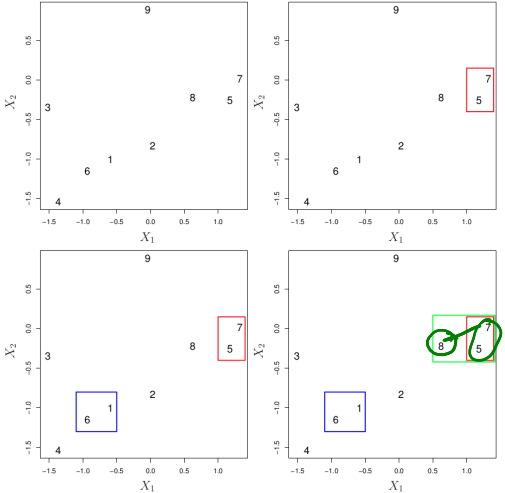


Distance between cluster A and cluster B:

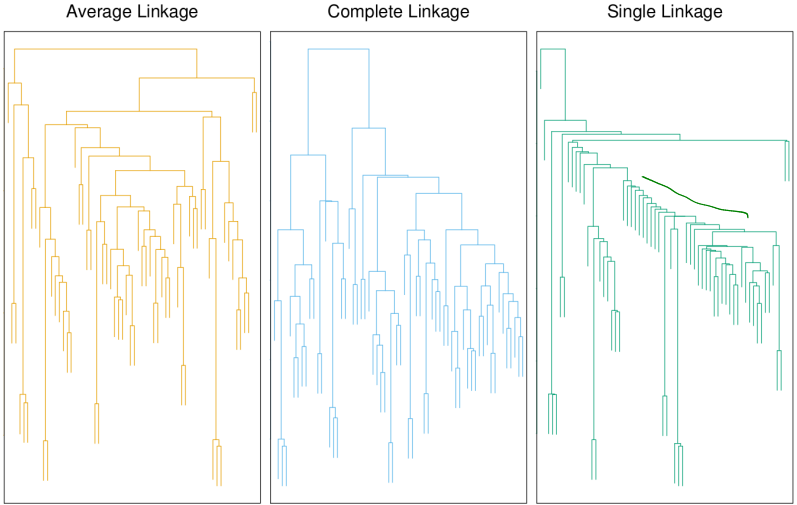
Largest distance between the points

$$L(A, B) = \max_{a \in A, b \in B} \|a - b\|$$

Complete



Examples of different linkage



Dependence on dissimilarity measure

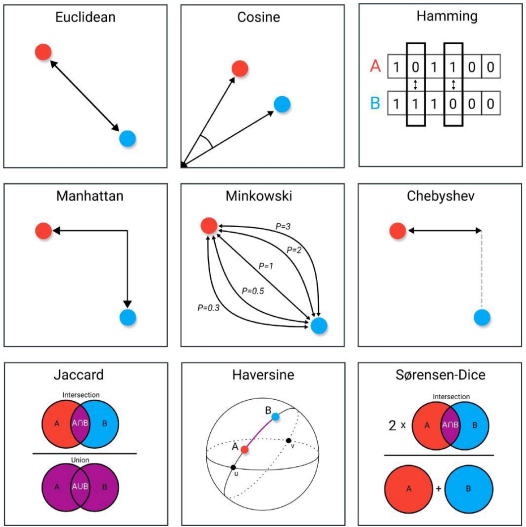
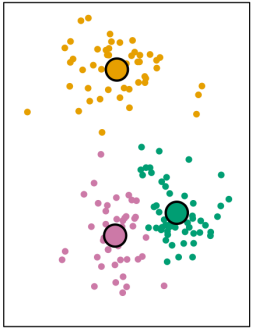


Photo Credit Link

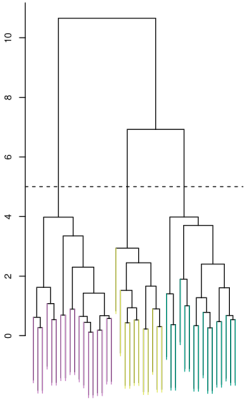
To the mini lab!!!

TL;DR

K-Means



Hierarchical clustering



Next time

Next time:

- More clustering
- Bring computer for some interactive mini-lab stuff

This week:

- Mon 4/25: 12.1, 12.4: Clustering
- Weds 4/27: Finish 12.4 & Mini lab
  - HW #10 Due Weds
- Fri 4/29: Review - Bring questions!
- Final: May 4, 5:45 - 7:45 PM, Erickson 226
  - Content since midterm (Ch 6 and on)
  - One page (8.5x11) handwritten cheat sheet